Chapter 3 • Sets and Functions

Solved Exercise 3.1

1. Write each set in set-builder notation.

(i) $\{1,4,9,16,25,36,\ldots,484\} = \{x|x = n^2, n\in N, 1\le n\le 22\}$.

(ii) $\{2,4,8,16,\ldots,256\} = \{x|x = 2^n, n\in N, 1\le n\le 8\}$.

(iii) $\{0,\pm1,\pm2,\ldots,\pm1000\}=\{x|x\in Z,-1000\le x\le 1000\}$.

(iv) $\{6,12,18,\ldots,120\} = \{x|x = 6n, n\in N, 1\le n\le 20\}$.

(v) $\{100,102,104,\ldots,400\} = \{x|x = 100+2n, n\in W, 0\le n\le 150\}$.

(vi) $\{1,3,9,27,81,\ldots\} = \{x|x = 3^n, n\in W\}$.

(vii) $\{1,2,4,5,10,20,25,50,100\}=\{x|x$ is a positive divisor of 100$\}$.

positive divisors of 100

(viii) $\{5,10,15,\ldots,100\} = \{x|x = 5n, n\in N, 1\le n\le 20\}$.

(ix) The integers strictly between -100 and 1000: $\{x|x\in Z,-100<x<1000\}$.

2. Write each set in tabular (roster) form.

(i) $\{x|x$ is a multiple of 3 and $x\le 36\}$: taking positive multiples as intended gives $\{3,6,9,\ldots,36\}$.

(ii) $\{x|x\in R$ and $2x+1=0\}$: $x=\frac{-1}{2}$, so the set is $\{\frac{-1}{2}\}$.

(iii) $\{x|x\in P$ and $x<12\}$, where P denotes primes: $\{2,3,5,7,11\}$.

(iv) $\{x|x$ is a positive divisor of 128$\}$: $\{1,2,4,8,16,32,64,128\}$.

(v) $\{x|x=2^n, n\in N$ and $n<8\}$: use $n=1,\ldots,7$ to get $\{2,4,8,16,32,64,128\}$.

(vi) $\{x|x\in N$ and $x+4=0\}$: $x=-4\notin N$, so the set is $\emptyset$.

(vii) $\{x|x\in N$ and $x=x\}$: every natural number satisfies the condition, giving $\{1,2,3,4,\ldots\}$.

N

(viii) $\{x|x\in Z$ and $3x+1=0\}$: $x=\frac{-1}{3}\notin Z$, so the set is $\emptyset$.

3. Write two proper subsets of each given set. Other correct examples are possible.

(i) For $\{a,b,c\}$, choose $\{a\}$ and $\{b,c\}$; both exclude at least one original element.

{a},{b,c}

(ii) For $\{0,1\}$, choose $\emptyset$ and $\{0\}$.

∅,{0}

(iii) For N, choose the even natural numbers and the odd natural numbers.

even naturals,odd naturals

(iv) For Z, choose N and the set of even integers.

N,even integers

(v) For Q, choose Z and $\{x\in Q|0<x<1\}$.

Z,{x∈Q|0<x<1}

(vi) For R, choose Q and the set of irrational numbers.

Q,irrationals

(vii) For $\{x\in Q|0<x\le 2\}$, choose $\{1\}$ and $\{x\in Q|0<x<1\}$.

{1},{x∈Q|0<x<1}.

4. Is there a set with no proper subset?

Yes. The empty set $\emptyset$ has only one subset, namely $\emptyset$ itself, so it has no proper subset.

5. Distinguish {a,b} and {{a,b}}.

$\{a,b\}$ has two elements, a and b. $\{\{a,b\}\}$ has one element, and that element is the set $\{a,b\}$.

6. Find the number of elements in each power set.

(i) The set is $\emptyset$, so $n=0$ and $|P(\emptyset)|=2^0=1$.

(ii) The set $\{0,1\}$ has $n=2$, so its power set has $2^2=4$ elements.

4

(iii) $\{1,2,3,4,5,6,7\}$ has $n=7$, so its power set has $2^7=128$ elements.

128

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(iv) {0,1,2,3,4,5,6,7} has n=8, so its power set has $2^8=256$ elements.
256
(v) {a,{b,c}} has two elements—a and the set {b,c}—so its power set has $2^2=4$ elements.
4
(vi) {{a,b},{b,c},{d,e}} has three set-valued elements, so its power set has $2^3=8$ elements.
8.

7. Write each power set.
(i) P({9,11}) = {∅, {9}, {11}, {9,11}}.
(ii) For S = {+, −, ×, ÷}: include ∅; four singletons; six pairs {+,−}, {+,×}, {+,÷}, {−,×}, {−,÷}, {×,÷}; four triples {+,−,×}, {+,−,÷}, {+,×,÷}, {−,×,÷}; and S. Total 16 subsets.
(iii) P({∅}) = {∅, {∅}}.
(iv) P({a,{b,c}}) = {∅, {a}, {{b,c}}, {a,{b,c}}}.

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